

Yuqin Duan (Sophia)

Curriculum Vitae

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EDUCATION

Massachusetts Institute of Technology Ph.D. in Electrical Engineering and Computer Science Advisor: Prof. Dirk R. Englund	Aug. 2021–Sept. 2026
Massachusetts Institute of Technology M.S. in Electrical Engineering and Computer Science	Aug. 2020–Aug. 2021
Purdue University B.S. in Electrical Engineering with distinction(top 1%)	Aug. 2015–May. 2019

EXPERIENCE

Massachusetts Institute of Technology , Postdoc Mentor: Prof. Polina Golland, Dr. Ellen Grant	Sept 2026–Sept 2028
NTT Research Inc. , Research Scientist Intern Mentor: Dr. Ryan Hamerly, Dr. Saumil Bandyopadhyay	May 2024–Aug 2024
Harvard Microrobotics Laboratory , Visiting Scientist Mentor: Prof. Kevin Yufeng Chen	Oct 2019–May 2020

PUBLICATIONS

- [1] **Y. Duan**, Q. Gu, H. Wang, Y. Hu, M. E. Trusheim, and D. R. Englund, “Ångström-accurate and precise localization microscopy”, in *Single Molecule Spectroscopy and Superresolution Imaging XIX*, International Society for Optics and Photonics, vol. 13858, SPIE, 2026.
- [2] **Y. Duan**, C. Espitia-Alvarez, Y. Hu, H. Wang, K. C. Chen, D. R. Englund, and M. E. Trusheim, “A machine learning based study of vertically loaded diamond microdisk resonator”, *CLEO*, May 2025.
- [3] **Y. Duan**, Q. Gu, J. Liu, M. E. Trusheim, and D. R. Englund, “Digital twin assisted end-to-end optical alignment system”, *arXiv*, 2025.
- [4] **Y. Duan**, Q. Gu, H. Wang, Y. Hu, K. C. Chen, M. E. Trusheim, and D. R. Englund, “A bayesian approach towards atomically-precise localization in fluorescence microscopy”, *Nature Communications*, vol. 16, no. 1, p. 9289, 2025.
- [5] H. Wang, S. Wu, K. Jacobs, **Y. Duan**, D. R. Englund, and M. E. Trusheim, “Cavity-enhanced solid-state nuclear spin gyroscope”, *Phys. Rev. Lett.*, vol. 134, p. 183 603, 18 Apr. 2025.
- [6] Y. Hu, S. Nagle, **Y. Duan**, H. Wang, and D. Englund, “Developing 3d models of atom-like defect spin memories in crystals for quantum technology research and education”, *Journal of Chemical Education*, vol. 101, no. 8, pp. 3522–3530, 2024.
- [7] H. Flores, B. Szamosfalvi, **Y. Duan**, I. Hammond, R. Camacho, and D. Englund, “Optimized free space emission from layered diamond microdisk resonators”, 2023 Intermountain Engineering, Technology and Computing (IETC), 2023.

- [8] **Y. Duan**, M. E. Trusheim, Y. Yu, K. C. Chen, and D. R. Englund, “Nanometer-scale fabrication and localization of quantum emitters in diamond”, MRS Spring, 2022.
- [9] **Y. Duan**, K. C. Chen, D. R. Englund, and M. E. Trusheim, “A vertically-loaded diamond microdisk resonator spin-photon interface”, *Opt. Express*, vol. 29, no. 26, pp. 43 082–43 090, Dec. 2021.
- [10] **Y. Duan**, K. C. Chen, M. E. Trusheim, and D. R. Englund, “A vertically loaded diamond microdisk resonator (vldmrt) towards a scalable quantum network”, CLEO: QELS Fundamental Science, 2021.
- [11] I. M. Hammond, **Y. Duan**, M. E. Trusheim, D. R. Englund, and R. M. Camacho, “Microdisk design for vertical collection from quantum emitters”, in *Frontiers in Optics + Laser Science 2021*, Optica Publishing Group, 2021, FTh4E.7.
- [12] J. Qin, H. Yan, G. Qiu, M. Si, P. Miao, **Y. Duan**, W. Shao, *et al.*, “Hybrid dual-channel phototransistor based on 1D t-se and 2D ReS 2 mixed-dimensional heterostructures”, *Nano Res.*, 2019.
- [13] M. Si, A. K. Saha, S. Gao, G. Qiu, J. Qin, **Y. Duan**, J. Jian, *et al.*, “A ferroelectric semiconductor field-effect transistor”, *Nat Electron*, 2019.
- [14] **Y. Duan**, E. Berger, R. Kandakatla, *et al.*, “The relationship between demographic characteristics and engagement in an undergraduate engineering online forum”, *2018 IEEE Frontiers*, 2018.
- [15] J. K. Qin, G. Qiu, W. He, J. Jian, M. W. Si, **Y. Duan**, *et al.*, “Epitaxial growth of 1D atomic chain based se nanoplates on monolayer ReS2 for High-Performance photodetectors”, *Adv. Funct. Mater.*, 2018.
- [16] M. Si, P.-Y. Liao, G. Qiu, **Y. Duan**, and P. D. Ye, “Ferroelectric Field-Effect transistors based on MoS2 and CuInP2S6 Two-Dimensional van der waals heterostructure”, *ACS Nano*, vol. 12, no. 7, pp. 6700–6705, Jul. 2018.

INVITED TALK

- [Mar 2026] **Stanford the D-Lab**, *Atomically Precise Imaging and Control of Nanophotonic Quantum Systems* (host: Prof. Jennifer Dionne)
- [Mar 2026] **MIT Medical Vision Group**, *Angstrom-Precise and Accurate Localization* (host: Prof. Polina Golland)
- [Sept 2025] **MIT Computational Biophotonics Lab**, *Atomic Fluorescent Localization Microscopy* (host: Prof. Sixian You)
- [Aug 2025] **Stanford Nanoscale and Quantum Photonics Lab**, *Precise Measurements in Quantum Technology* (host: Prof. Jelena Vuckovic)
- [May 2025] **Stanford Computational Imaging Lab**, *A Bayesian Approach to Atomic Localization Fluorescent Microscopy* (host: Prof. Gordon Wetzstein)
- [Mar 2025] **Boston University Photonics Center Seminar Series**, *A Bayesian Approach to Atomic Localization Fluorescent Microscopy* (host: Prof. Tianyu Wang)

TEACHING

- **Instructor** of 6.S094 Intro to Quantum Network
First-time offered in 2023: sites.mit.edu/sophiayueqin/mit-6s094-iap-2023/
- **Teaching Assistant** at Purdue University
ECE 33700(ASIC Design), ECE 20200(Linear Circuit II)

PROFESSIONAL ACTIVITIES

- **President** of MIT’s Interdisciplinary Quantum Information Science and Engineering (iQuISE) 2024-2025

- Organized with MIT Center of Quantum Engineering(CQE)*
- **Co-Chair** of 1st, 2nd, 3rd Quantum Annual Research Conference(QuARC) 2022-2024
Organized with MIT Center of Quantum Engineering(CQE)
- **Blog Writer** at MIT
- **Ambassador** at Birck Nanotechnology Center

MENTOR

- **MIT Undergraduate Research Opportunities Program (UROP):** 2023-present
Armando Martinez de la Villaz, now PhD at MIT
Adrian Kuka, now at MITRE
Hanming Xiong, now Master at NYU
Kevin R. Palma, Leo K.N. Kana, Christopher G Espitia-Alvarez, now undergrad at MIT
- **Research Science Institute (RSI):** 2022-2024
Ryan Razon, now undergrad at Harvard
Jerry Liu, now undergrad at MIT, winning top 5 best paper
Amir Almukhanov, now undergrad at MIT

SCHOLARSHIPS AND AWARDS

- Mathworks Fellowship 2023–2024
- Edwin S Webster Grad Fellowship 2019–2020
- Dean’s List Semester Honor 2015–2019
- Eta Kappa Nu Outstanding Scholarship 2018–2019
- RCA Zworykin ECE Scholarship 2017–2018
- Hackathon BoilermakerIV Qualcomm Smart Home Prize 2019
- Hackathon BoilermakerV Rockwell Collins: Best Use of Communication Prize 2018
- Hackathon BoilermakerIV Qualcomm Best IoT Prize 2017

EXTRACURRICULAR ACTIVITIES

- **Senior Choreographer** at MIT ADT 2019–2022
Senior traditional and contemporary choreographer
- **Co-founder** of Science in Melody 2020–2022
A youtube channel connects scientists with music
- **Coffee Hour Chair** at MIT Sid-Pac 2020–2021
Organized theme events every Wednesday evening
- **Principle Dancer** at Purdue PCPAT 2019–2020
Lead and help to teach the performance for culture showcase
- **Learning Committee Chair** at Purdue IEEE 2016–2017
Organized technical workshops, professor talks and research fair

SKILLS

- **Skills:** FDTD (Lumerical, tidy3D), COMSOL, Microscopy, nanofabrication, gds Spy, PyTorch, chip packaging
- **Language:** Mandarin(native), English(professional), German(conversational), Spanish(conversational)
- **Hobbies:** I dance with the ocean’s waves, the mountain’s snow, and the sunset’s glow